1.8 Asymptotic Notations

Big-O, Big-Ω and Big-Θ — Complete Notes based on Abdul Bari Lectures

1.8.1 Asymptotic Notations — Big-O, Big-Ω and Big-Θ

1. Introduction to Asymptotic Notations

Asymptotic notations originate from mathematics. They provide a concise, communicable way to express the time complexity and growth class of algorithms without having to write out lengthy functions every time.

Core Notations:

- **Big-O (O):** Upper bound of a function.
- **Big-Omega (Ω):** Lower bound of a function.
- **Big-Theta (Θ):** Tight bound of a function.

2. Big-O Notation (O) — Upper Bound

Definition: f(n) = O(g(n)) if and only if there exist positive constants c and n■ such that:

$$f(n) \leq c \cdot g(n), \quad \text{for all } n \geq n_■$$

Example: For f(n) = 2n + 3:

$$2n + 3 \leq 2n + 3n = 5n, \quad \text{for all } n \geq 1$$

Here, c = 5 and g(n) = n, so **f(n) = O(n)**.

Important: Higher-order functions like O(n²) or O(2■) are mathematically also valid upper bounds, but they are not useful. We should choose the closest upper bound, O(n). A lower-growing function such as O(log n) cannot serve as an upper bound here.

3. Big Omega Notation (Ω) — Lower Bound

Definition: f(n) = Ω(g(n)) if and only if there exist positive constants c and n■ such that:

$$f(n) \geq c \cdot g(n), \quad \text{for all } n \geq n_■$$

Example: For f(n) = 2n + 3:

$$2n + 3 \geq 1n, \quad \text{for all } n \geq 1$$

Here, c = 1 and g(n) = n, so **f(n) = Ω(n)**.

Important: Smaller functions like Ω(log n) are mathematically valid lower bounds, but the nearest lower bound Ω(n) is the most useful. Higher-order functions like Ω(n²) cannot be lower bounds here.

4. Theta Notation (Θ) — Tight Bound

Definition: f(n) = Θ(g(n)) if and only if there exist positive constants c■, c■, and n■ such that:

$$c_■ \cdot g(n) \leq f(n) \leq c_■ \cdot g(n), \quad \text{for all } n \geq n_■$$

Example: Combining the bounds from f(n) = 2n + 3:

$$1n \leq 2n + 3 \leq 5n, \quad \text{for all } n \geq 1$$

Since g(n) = n on both sides, **f(n) = Θ(n)**. Θ notation is preferred whenever possible because it represents the exact/tight rate of growth.

5. Important Distinction: Bounds vs. Cases

Common Misconception: Big-O is not synonymous with worst-case, nor is Omega synonymous with best-case.

Clarification: Best-case, worst-case, and average-case describe algorithmic input scenarios, whereas O, Ω, and Θ are mathematical bounds that can be applied to any case.

1.8.2 Asymptotic Notations — Examples

1. Example 1: Polynomial Function

Given Function: f(n) = 2n² + 3n + 4

Finding the Upper Bound (O):

Convert all terms to the highest-degree term:

$$2n^2 + 3n + 4 \leq 2n^2 + 3n^2 + 4n^2 = 9n^2, \quad \text{for all } n \geq 1$$

With c = 9 and g(n) = n², we get: **f(n) = O(n²)**.

Finding the Lower Bound (Ω):

$$2n^2 + 3n + 4 \geq 1n^2, \quad \text{for all } n \geq 1$$

Therefore: **f(n) = Ω(n²)**.

Tight Bound (Θ):

$$1n^2 \leq 2n^2 + 3n + 4 \leq 9n^2$$

Since the function is bounded on both sides by n²: **f(n) = Θ(n²)**.

2. Example 2: Log-Linear Combination

Given Function: f(n) = n² log n + n

Bounding the Function:

Upper Bound: $n^2 \log n + n \leq 10n^2 \log n \Rightarrow$ **$f(n) = O(n^2 \log n)$**.

Lower Bound: $n^2 \log n + n \geq 1n^2 \log n \Rightarrow$ **$f(n) = \Omega(n^2 \log n)$**.

Tight Bound: Since both bounds match: **$f(n) = \Theta(n^2 \log n)$**.

Growth Order Position:

$$n^2 < n^2 \log n < n^3$$

Therefore, $n^2 \log n$ lies between quadratic and cubic growth.

3. Example 3: Factorial Function (n!)

Expansion: $n! = 1 \times 2 \times 3 \times \blacksquare \times n$

Upper Bound: Replace every factor with n:

$$n! \leq n \times n \times \blacksquare \times n = n\blacksquare$$

Therefore: **$n! = O(n\blacksquare)$**.

Lower Bound: Replace every factor with 1:

$$n! \geq 1 \times 1 \times \blacksquare \times 1 = 1$$

Therefore: **$n! = \Omega(1)$**.

Why Upper/Lower bounds are needed here: The upper bound is n■ and the trivial lower bound is 1. Because the bounding functions do not match, a simple tight bound (Θ) is not directly formed by these bounds. This illustrates why O and Ω are essential when an exact tight class isn't simple to pin down.

4. Example 4: Log Factorial (log n!)

Expansion: $\log(n!) = \log(1 \times 2 \times 3 \times \blacksquare \times n)$

Upper Bound:

$$\log(n!) \leq \log(n \times n \times \blacksquare \times n) = \log(n\blacksquare) = n \log n$$

Therefore: **$\log(n!) = O(n \log n)$**.

Lower Bound: The simple argument gives $\log(n!) \geq \log(1 \times 1 \times \blacksquare \times 1) = \log 1 = 0$ (or a trivial $\Omega(1)$ bound).

5. Key Takeaways & Practical Analogy

Preference for Θ: Whenever possible, use Θ notation because it guarantees a tight rate of growth.

Choosing Useful Bounds: If using O, choose the tightest upper bound (e.g., if a function is n^2, calling it $O(n^2)$ is informative; calling it $O(n^3)$ is mathematically true but unhelpful). If using Ω, choose the highest/closest lower bound.

Phone Price Analogy: Estimating a phone's price as ~■20,000 gives the exact target (Θ). Saying “at least ■2,000” (Ω) or “at most ■2,00,000” (O) is technically correct, but too wide to be practically useful unless tightened.

■ Quick Memory

$O \rightarrow$ Upper | $\Omega \rightarrow$ Lower | $\Theta \rightarrow$ Tight

Best/Worst/Average Case $\neq$ O/Ω/Θ

These notes follow the Abdul Bari 1.8.1 and 1.8.2 material provided.